



PROJECT REQUIREMENTS- PORTFOLIO BUDDY



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Inhalt

Project Introduction	2
Planned Features (as User Stories).....	2
Must-Haves:	2
Nice-To-Haves:	4
Pages	5
1.Main Page (Public):.....	5
2. Registration/Sign-Up Page (Public):.....	5
3. Login Page (Public):.....	5
4. User Dashboard (Private – Users):	5
5. Add/Edit Asset Page (Private – Users):.....	5
6. Profile Page (Private – Users):.....	5
7. Admin Dashboard (Private – Admins)	5
File Upload Use	6
Asset Attachments	6
Profile Pictures.....	6
Access Restrictions	7
ChangeLog.....	8

Project Introduction

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PortfolioBuddy is an online financial portfolio management tool aimed at assisting users in monitoring and evaluating their investments all in one location. The primary concept is to offer users a centralized interface for all their assets, such as stocks, commodities like gold, and funds in bank accounts that accrue interest, enabling them to effortlessly track their overall net worth and investment results.

The app aims to ease portfolio management by automatically retrieving current financial data and determining the overall value of the portfolio in real time. This enables users to make educated financial choices and quickly grasp their asset distribution. The intended users include individual investors and savers seeking a simple, accessible method to handle different assets (ranging from novices managing personal savings to seasoned investors overseeing various asset types).

Planned Features (as User Stories)

Must-Haves:

1. As a new user, I want to create an account by registering with my email and a password, so that I can have my own profile and start using the portfolio management features. (Registration/Sign-Up Page (Public))
has been solved in ***src/pages/register.php lines 1-209*** (Full controller logic for POST handling, hashing, DB insertion, and View) and ***src/pages/util/register_functions.php lines 1-29*** (Validation orchestration). It also utilizes core validation logic in ***src/pages/util/input_validation_function.php lines 1-57***
2. As a registered user, I want to log into the website using my credentials, so that I can access my personal portfolio dashboard securely. (Login Page (Public))
has been solved in ***src/pages/login.php lines 1-136*** (Session handling, DB verification of hash, and View) and ***src/pages/util/login_functions.php lines 1-19*** (Input normalization and validation). It also utilizes core validation logic in ***src/pages/util/input_validation_function.php lines 1-31***
3. As a logged-in user, I want to view and update my profile information (including uploading a profile picture), so that I can personalize my experience and keep my account details current. (Profile Page (Private – Users))
has been solved in ***src/pages/profile.php lines 1-158*** (Controller logic for updates, file handling for image upload in ***lines 38-49***, and View), ***src/pages/util/profile_functions.php lines 1-31*** (Input validation), and ***src/pages/util/user_persistence_functions.php lines 6-42*** (Database update logic). The profile image display logic is handled in ***src/pages/includes/_navbar.php lines 4-24***.
4. As a user, I want to add assets(must have=shares) to my portfolio, so that I can track all my investments in one place. (Add/Edit Asset Page (Private – Users))
has been solved in ***the src/pages/positions.php*** file especially: ***lines 93-156*** (INSERT/Update(ticker) logic), ***lines 170-216*** edit and load asset functionalities, ***lines 282-329*** (Input Form). It relies on

src/pages/util/positions_functions.php lines 7-99 (Input validation) and *src/pages/util/stock_data_functions.php lines 28-57* (Fetching Ticker from ISIN, needs the code **1-28** above to translate json into php array).

5. As a user, I want to see an overview of my entire portfolio's total value and its breakdown by asset classes, so that I understand my current net worth and asset allocation at a glance. (User Dashboard (Private – Users))

has been solved in *src/pages/dashboard.php lines 18-40* (Data fetching and total value(of individual share not the networth- extra helper) calculation), **57-73** Page Introduction/User Welcomeing, *lines 130-155* (Chart.js configuration for breakdown), **and lines 74-120** (Tabular overview).

src/pages/util/dashboard_metrics.php lines 6-15 contains the helper function `getNetWorth()`

6. As a user, I want to be able to update asset details (for example, if I bought more of a stock or if an interest rate changed) or remove an asset from my portfolio, so that my portfolio data remains accurate and reflects my real holdings. (Add/Edit Asset Page (Private – Users))

has been solved in *src/pages/positions.php* lines **62-74** delete assets, *lines 94-216* insert, edit and load asset functionalities, *lines 282-329* (Input Form). It relies on *src/pages/util/positions_functions.php lines 7-99* (Input validation) and *src/pages/util/stock_data_functions.php lines 28-57* (Fetching Ticker from ISIN, needs the code **1-28** above to translate json into php array).

7. As a user, I want to upload and attach a document or image to a particular asset so that I can keep important asset-related documents organized within the application. Add/Edit Asset Page (Private – Users))

has been solved in *src/pages/positions.php* lines 158-166 (File move/upload logic), lines 38-60 (File download logic), lines 76-92 (File deletion logic), and lines 236-261 (Document list view). Supporting logic is in *src/pages/util/positions_functions.php* lines 68-84 (MIME type and size validation) and lines 127-160 (Directory management and listing).

8. As an administrator, I want to access an admin panel where I can view all user accounts, update their details, or deactivate/delete accounts if needed, so that I can manage the platform's users. (Admin Dashboard (Private – Admins))

has been solved in *src/pages/adminpanel.php lines 67-80* (Fetching all users from DB), *lines 115-177* (Table View rendering user list and actions), and *lines 29-62* (Controller logic for handling role change, delete and password reset actions). It relies on *src/pages/util/user_persistence_functions.php lines 96-113* (delete_user_account logic) and *lines 47-64* (update_user_password logic used for resets).

9. As an administrator, I want to be able to assign or change user, so that I can delegate administrative responsibilities when necessary. (Admin Dashboard (Private – Admins))

has been solved again in *lines 29-62* (Controller logic for handling role change, delete and password reset actions), specifically in

src/pages/adminpanel.php lines 39-45 (Specific controller logic to handle 'set_role' POST requests) and **lines 141-157** (Form and Button implementation to toggle roles). It utilizes the core logic in **src/pages/util/user_persistence_functions.php lines 69-91** (update_user_role function).

Nice-To-Haves(=Not a must have):

1. As a user, I want the application to automatically fetch the latest market prices for my assets (for example, stock prices, gold prices, and interest updates on cash accounts) via financial APIs, so that the portfolio values and charts are always up-to-date.
Not finished, development has been started in **stock_data_functions.php lines 66-94**, code snippet to fetch historical data, not utilized so far.
2. As a user, I want to click on or select a specific asset or asset class to see detailed information (such as its individual performance chart, purchase history, and any related documents), so that I can analyze specific parts of my portfolio in depth.
Not finished, development has been started in **stock_data_functions.php lines 66-94**, code snippet to fetch historical data, not utilized so far
3. As a user, I want to view charts of my portfolio's performance over time (overall and by asset category), so that I can easily visualize trends and the growth or decline of my investments.
Not finished, development has been started in **stock_data_functions.php lines 66-94**, code snippet to fetch historical data, not utilized so far
4. As a user, I want to compare my portfolio's performance against benchmarks like the S&P 500 or MSCI World, so that I know whether I'm underperforming or outperforming the market.
5. As a user, I want to see a breakdown of my ETF holdings into their underlying stocks, so that I understand which companies and sectors I'm really invested in.

Pages

1. **Main Page (Public):** The entry page of the application. It will present the features and advantages of PortfolioBuddy, along with general details about the application. This page is expected to feature a greeting and a summary of the app's functionality (something like: "Manage all your investments in one location")
2. **Registration/Sign-Up Page (Public):** The site where new users have the ability to establish an account. It includes a registration form that gathers information such as name, email(= username), and password
3. **Login Page (Public):** The page for existing users to log into their account. It contains a login form where users enter their email/username and password. After submitting, the backend (PHP) will verify the credentials against the database.
4. **User Dashboard (Private – Users):** This is the primary portfolio summary page displayed to a user upon logging in. It's a main dashboard showing the summary of the user's portfolio. Essential components on this page consist of:
 - the overall portfolio value (combining all assets)
 - a division of the portfolio by asset.
 - An overview list or table of the user's assets will be included, featuring important details.
 - An interface where the user can move to add a new asset, access detailed pages for each asset, or visit other sections such as the profile.
5. **Add/Edit Asset Page (Private – Users):** A section with a form that enables the user to enter information for a new asset or change details of an existing asset. To include an asset, the form will generally require information like asset, name, quantity, purchase price and any other details (as an example: for cash accounts(nice to have), the interest rate may be indicated). The user is permitted to upload a file on this page that is relevant to the asset.
6. **Profile Page (Private – Users):** The section that allows a logged-in user to see and modify their personal profile. It will display the user's details including name , email, and profile image. The user is able to modify their information here. For instance, upload a new profile picture or change their password. The page will feature a form for making those updates.
7. **Admin Dashboard (Private – Admins):** A dedicated dashboard page exclusively available to admin accounts. The page offers a summary of administrative details.

This page enables the administrator to view a list of all user accounts within the system. It could be a table containing users with details such as username, email, registration date, role (user or admin), and possible actions. From this point, the admin has the ability to execute actions like: view a user's information, modify a user's role, reset a user's password, or deactivate/delete a user's account

File Upload Use

File types supported: The platform is going to allow uploads of JPG images and PDF files.

Asset Attachments: Users are able to link files to their assets to enhance the organization of the assets. This enables crucial documents (such as receipts, statements, or certificates) to be kept together with the associated asset in the application.

Profile Pictures: Besides asset documents, users can upload a profile picture (in jpg file format) to customize their account.

Access Restrictions

Public pages don't have any restrictions, private pages are restricted to user's or admins.

If a page is restricted to users, the user has to have an active account and be logged-in (could also be an admin).

If a page is restricted to admins, only logged-in active admin accounts are able to view the page.

1. Main Page (Public)
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ChangeLog

Date	Changes	Editor
15.09.2025	Initial Version	Philip Stipkovits, Benjamin Hirsch
27.11.2025	Specified shares as the only must have asset and added resolving clarifications in page sections	Philip Stipkovits, Benjamin Hirsch